

MINI PROJECT – DAY 8

Hospital Data Pipeline

STEP 1 - Linux Setup

mkdir hospital_pipeline

cd hospital_pipeline

mkdir raw processed reports archive

touch raw/patients.csv raw/doctors.csv raw/appointments.csv

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ tree
.
├── archive
├── mini_projects.ipynb
├── processed
├── raw
│   ├── appointments.csv
│   ├── doctors.csv
│   └── patients.csv
└── reports

5 directories, 4 files
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$
```

STEP 2 - Add raw data

1. cat to view full contents

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat patients.csv | column -t -s,
patient_id  name          age  blood_type  city
1           Raj Kumar     34   O+          Delhi
2           Sneha Patel   39   A+          Mumbai
3           Tom Williams 46   B-          Chicago
4           Fatima Al-Sayed 29  AB+         Dubai
5           Liu Yang     24   O-          Beijing
6           Ana Gonzalez 56   A-          Madrid
7           James Okafor 32   B+          Lagos
8           Emily Carter 19   O+          Toronto
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat doctors.csv | column -t -s,
doctor_id  name          specialty          experience_years
1          Dr. Anil Mehta Cardiology         15
2          Dr. Sarah Johnson Neurology          12
3          Dr. Raj Singh   Orthopedics        8
4          Dr. Lisa Park   Dermatology         6
5          Dr. Carlos Torres General Medicine    20
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat appointments.csv | column -t -s,
appt_id  patient_id  doctor_id  appt_date  diagnosis              fee
1         1           1          2024-01-10 Hypertension checkup    500
2         2           5          2024-01-11 Fever and cold           200
3         3           3          2024-01-12 Knee pain evaluation    750
4         4           2          2024-01-13 Migraine treatment      600
5         5           4          2024-01-14 Acne treatment          300
6         1           5          2024-01-15 Follow-up general checkup 200
7         6           1          2024-01-16 Cardiac stress test      900
8         7           3          2024-01-17 Back pain physiotherapy  400
9         8           5          2024-01-18 Routine vaccination     150
10        2           2          2024-01-19 Epilepsy management     850
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

STEP 3 - Linux Profiling (do this before Python)

For each of the three CSV files, run:

1. wc -l to count rows

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat patients.csv | wc -l
9
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat doctors.csv | wc -l
6
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cat appointments.csv | wc -l
11
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

2. head -1 to confirm headers

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ head -1 patients.csv
patient_id,name,age,blood_type,city
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ head -1 doctors.csv
doctor_id,name,specialty,experience_years
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ head -1 appointments.csv
appt_id,patient_id,doctor_id,appt_date,diagnosis,fee
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

For appointments.csv:

4. Extract and count unique doctor_ids

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cut -d',' -f3 appointments.csv | sort | uniq -c
 2 1
 2 2
 2 3
 1 4
 3 5
 1 doctor_id
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

5. Find the appointment with highest fee

using sort + head

```
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ sort -t',' -k6 -rn appointments.csv | head -n 1
7,6,1,2024-01-16,Cardiac stress test,900
prashant@DESKTOP-B9KE2N7:/mnt/d/Kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ |
```

STEP 4 - Python Profiling Script

Create: raw/profiler.py

Write code that:

1. Reads all three CSV files using pandas.
2. For each file prints:
 - Filename
 - Shape (rows x columns)
 - Column names
 - Data types
 - Count of NULL values per column
 - First 3 rows
3. For appointments specifically:
 - Total fee collected
 - Average fee
 - Min and max fee
 - Count of appointments per doctor_id

```
...  
  
def code_divider():  
    print("-----")  
  
# 1. Reads all three CSV files using pandas.  
  
import pandas as pd  
  
def profile(df, name):  
    code_divider()  
    print(f"Filename: {name}")  
  
    print(f"Shape: {df.shape[0]} rows x {df.shape[1]} columns")  
  
    print("\nColumns:")  
    print(list(df.columns))  
  
    print("\nData Types:")  
    print(df.dtypes)  
  
    print("\nNull Values:")  
    print(df.isnull().sum())  
  
    print("\nFirst 3 Rows:")  
    print(df.head(3))
```

```

def main():
    patients = pd.read_csv("patients.csv")
    doctors = pd.read_csv("doctors.csv")
    appointment = pd.read_csv("appointments.csv")

    profile(patients, "Patients")
    profile(doctors, "Doctors")
    profile(appointment, "Appointments")

    code_divider()
    print("Appointments Analysis")

    total_fee = appointment["fee"].sum()
    avg_fee = appointment["fee"].mean()
    min_fee = appointment["fee"].min()
    max_fee = appointment["fee"].max()

    print(f"Total Fee Collected: {total_fee}")
    print(f"Average Fee: {avg_fee}")
    print(f"Minumum Fee: {min_fee}")
    print(f"Maximum Fee: {max_fee}")

    doctor_counts = appointment.groupby("doctor_id").size()

    print("\nAppointments per Doctor:")
    print(doctor_counts)

    top_doctor = doctor_counts.idxmax()
    top_visits = doctor_counts.max()

    code_divider()
    print("Hospital Pipeline - Day 8")

    print(f"Patients : {len(patients)} records")
    print(f"Doctors : {len(doctors)} records")
    print(f"Appointments: {len(appointment)} records")
    print(f"Total Revenue: {total_fee}")
    print(f"Avg Fee      : {round(avg_fee, 2)}")
    print(f"Top Doctor by Appointments: doctor_id {top_doctor} ({top_visits} visits)")

if __name__ == "__main__":
    main()

```

Output:-

```

-----
Filename: Patients
Shape: 8 rows x 5 columns

Columns:
['patient_id', 'name', 'age', 'blood_type', 'city']

Data Types:
patient_id    int64
name          object
age           int64
blood_type    object
city          object
dtype: object

Null Values:
patient_id    0
name          0
age           0
blood_type    0
city          0
dtype: int64

First 3 Rows:
   patient_id      name  age blood_type    city
...
Appointments: 10 records
Total Revenue: 4850
Avg Fee      : 485.0
Top Doctor by Appointments: doctor_id 5 (3 visits)

```